B290 ROOFING

Pine Ridge Job Corps CCC

Specification List

Section	Section Name
010150	General Requirements
017329	Cutting and Patching
024119	Selective Demolition
061600	Rough Carpentry
075419	Polyvinyl-Chloride Roofing
076200	Sheet Metal Flashing and Trim
077100	Roof Specialties

SECTION 010150
GENERAL REQUIREMENTS

PART 1 – SUMMARY OF WORK

1.1 DESCRIPTION

A. Work for this project requires the contractor to be responsible for supplying all labor, materials, equipment, utilities, miscellaneous items, freight and delivery and all necessary valid licenses for the work summarized as follows:

1. Remove indicated roof gutter systems and replace with new gutter systems. Install Class A, B, or C fire rated roof assembly that meets wind pressure requirements over existing metal roof system with 20-year No Dollar Limit (NDL) warranty. New roof system shall consist of polyisocyanurate flute fill insulation and continuous insulation with a PVC membrane. Replace or extend gas and plumbing vents and roof curbs to provide required clearance above new roof system. Provide all (counter)-flashings, terminations, fasteners, sealants, and other accessories required for a weathertight installation conforming to the requirements of the referenced codes, standards, and manufacturer installation instructions and requirements for NDL warranty. Existing metal roof over outside storage areas and entrance canopy will remain as is; provide transition between new and existing roofing systems at these locations. Basis of Design (BOD) roofing system by Durolast “Metal Retrofit” is specified; substitutions shall meet or exceed all salient features of the BOD system in order to be approved. Provide all inspection services and manufacturer’s inspections required for project warranty.

1.2 LOCATION

A. This project is located at the Pine Ridge Job Corps Civilian Conservation Center located twelve miles from Chadron, Nebraska in Dawes County. The address is 15710 Highway 385, Chadron, NE 69337 . Driving instructions can be obtained from the Center by calling 308-432-3316.

1.3 WORK HOURS

A. Work hours for this project are between 8:00 AM-4:30 PM Monday through Friday.

1.4 USE OF PREMISES

A. Confine constructions operations to immediate area of work. Do not disturb portions of Project site beyond areas in which the Work is indicated.

B. Government will occupy site and buildings during entire construction period. Cooperate with Government during construction operations to minimize conflicts and facilitate

Government usage. Perform the work so as not to interfere with Government's day-to-day operations.

C. Keep driveways, parking areas, loading areas and entrances to buildings clear and available to Government, Government's employees, and emergency vehicles at all times. Do not use these areas for parking or storage of materials. Do not close or obstruct walkways, corridors, or other occupied or used facilities without written permission from Government.

1.5 DEFINITIONS

A. The following terms in any combination relating to the installation of products/materials: install, furnish, provide, or similar statements; mean that the contractor is responsible for the procurement and installation of the products unless it is specifically designated as either a government furnished item, supply only product, or spare.

1.6 INCIDENTAL CONSTRUCTION

A. Performance of the specific tasks indicated in the plans and specifications may require incidental construction including that is not specifically included in drawings and specification but not limited to: provision of temporary shoring or supports, installation and subsequent removal of temporary access points, removal and reinstallation or relocation of items not designated for replacement as required for work or equipment clearance or as a precautionary protective measure. Disconnection of supporting or interconnection equipment between different systems and reconnection to new equipment as applicable to specific project requirements; examples include but are not limited to: fire alarm components and interconnection/control for HVAC air handling equipment, cleaning chemical equipment for dishwashers, electric bonding for conductive water piping replacement, etc. In all cases the contractor shall be responsible for removal of temporary works, reinstallation of temporarily relocated items, or reconstruction of items selectively demolished for incidental construction purposes in a neat workman like manner, that matches existing finishes, quality standards, and code requirements unless the Contracting Officer specifically authorizes incidental construction to be incorporated into the finished work or items removed for coordination purposes to not be re-installed or reconstructed. The Contractor shall provide turn-key fully operational and finished installations unless the drawings and specifications specifically indicate otherwise.

1.7 FIELD VERIFICATION AND COORDINATION OF WORK

A. Drawings are schematic or typical in nature or intended to identify location of work. The Contractor shall be responsible for field verification of existing conditions and coordination of new work with those existing conditions. Examples below are intended to demonstrate the intent of this statement but are not inclusive of all field verification and coordination. Notify COR when specified requirements conflict with field verification.

1. General Field Verification: Verify existing construction finishes, locations of structural members that may obstruct work, and actual dimensions/distances for required work.
2. Electrical Field Verification: Inspect and test building voltage prior to ordering electrically operated equipment.
3. HVAC Field Verification: Verify space constraints of locations indicated for new equipment and equipment replacement and verify that submitted products will fit within space constraints with any code or manufacturer clearance requirements.
4. Installation Coordination: Install equipment and other components in accordance with referenced codes, standards, and manufacturer installation instructions. Do not obstruct electrical components or other equipment requiring routine service or their working clearances with new or replacement work; offset new work or relocate items requiring clearance to avoid conflicts between new and existing work. Offset new work to avoid major structural element such as beams, posts, columns, unless specifications or drawings direct modifications to those structural elements. Coordinate repair of cutting, patching, sealing or other disturbances of finish systems with surrounding areas. Disturbances to exterior walls or roofs shall be sealed with compatible/matching products, weatherproofed, and include thermal insulation and weather resistive barriers as applicable to existing building system. Disturbances to interior finishes shall be matched in a clean workmanlike manner consistent with the space type; public or employee work areas shall be neatly finished with high level finish system, mechanical/electrical rooms and similar spaces accessed solely for maintenance purpose have a lower class of aesthetic expectation but must be repaired to maintain the existing level of fire and smoke performance.

1.8 STANDARDS AND CODES

A. All contract work shall adhere to the most recent version of:
1. All the International Code Council Family of codes (ICC) e.g. International Building Code (IBC), International Fire Code (IFC), and International Mechanical Code (IMC).
2. All National Fire Protection Association (NFPA) codes and Standards e.g. , NFPA 70: National Electrical Code (NEC), NFPA 1: Fire Code, and NFPA 101- Life Safety Code.
3. Architectural Barriers Act Standards (ABA or ABAAS).
4. Product Manufacturer’s installation instructions.

1.9 QUALIFICATIONS

A. When applicable to specific contract work items:

1. Electrical work shall be performed by journey level electricians licensed in the State of contract performance or trainees/apprentices working under the direct supervision of State-licensed electrical supervisors (or State equivalent).
2. Plumbing work shall be performed by journey level plumbers licensed in the State of contract performance or trainees/apprentices working under the direct supervision of State-licensed plumbing supervisors (or State equivalent).

1.10 WORK RESTRICTIONS

A. Smoking is permitted on campus only in the designated smoking area.

1.11 SCHEDULE OF ITEMS

A. Schedule of Items:

1. Provide pricing for all items listed in the Schedule of Items.
2. Contract award will be made for all Base Items.
3. Optional Items may or may not be awarded, depending on funding availability.

B. Base Items (BI):

1. ***Base Item No. 1 –B290 Roofing***

a. Measurement: This item is measured as a Lump Sum quantity.
b. Payment: The contract lump sum price as shown in the Schedule of Items includes all materials, equipment, labor and incidentals to replace roof gutter systems and install insulated PVC roofing system (metal retrofit) on areas indicated and all other coordination work and incidental requirements necessary for a complete, weathertight roof.
c. Deficiencies: #152598 and 120616.

PART 2 – PROJECT SCHEDULE

2.1 PROJECT SCHEDULE

ITEM*	REQUIRED DUE DATE (calendar days)
Pre-Construction Conference	Within 10 days of Award (virtually)

Bonds	10 days after Award
Notice To Proceed (NTP)	Issued after Pre-Construction Conference (insurance and bonds acceptance required prior to NTP)
Action and Informational Submittals	12 days after NTP (CORs have 12 days to return)
Progress Reports	Every 7 days after NTP is issued.
Closeout Submittals	Prior to CO acceptance of project
Contract Release form and final pay request	10 days after CO acceptance of project

* All items above must be included in Contractor's construction schedule

PART 3 – SUBMITTAL PROCEDURES

3.1 DEFINITIONS

A. Action Submittals: Written and graphic information that requires Government approval.

B. Informational Submittals: Written information that does not require Government approval. Submittals may be rejected for not complying with requirements.

3.2 PROCEDURES

A. Contractor Responsibilities
   1. Review each submittal and check for compliance with the Contract Documents.
   2. Highlight, encircle, or otherwise identify deviations from the Contract Documents on submittals.
   3. Include the following information on label or cover sheet for processing and recording action taken:
      a. Project name.
      b. Date.
      c. Name and address of Contractor.
      d. Name of manufacturer.
      e. Unique identifier, including revision number.
      f. Number and title of appropriate Specification Section.

g. Drawing number and detail references, as appropriate.
h. Date of Contractor's approval.
i. Statement certifying that submittal has been reviewed, checked, and approved by the Contractor for compliance with the Contract Documents.

4. The Government will not review submittals that do not bear Contractor's approval and will return them without action.
5. Unless additional copies are required for final submittal, and unless the Contracting Officer (CO) observes noncompliance with provisions of the Contract Documents, initial submittal may serve as final submittal.
6. Submittals must be submitted in electronic PDF format. Submit to CO via email or electronic file transfer site.
7. Only final submittals with mark indicating action taken by Government are to be used for construction.
8. Submittals not required by the Contract Documents will not be reviewed and may be discarded.

B. Processing Time: Time for review shall commence on CO's receipt of submittal.
1. Initial Review: Government will review initial submittal within 12 calendar days. Additional time may be needed if processing must be delayed to permit coordination with subsequent submittals. CO will advise Contractor when a submittal being processed must be delayed for coordination.
2. Resubmittals: If resubmittal is necessary, it will be resubmitted within 3 calendar days, and reviewed by the Government within 3 calendar days.

3.3 ACTION SUBMITTALS

A. Prepare and submit Action Submittals required in the Technical Specifications in this contract, if applicable, and as indicated at the end of this subsection.

B. Collect information into a single submittal for each element of construction and type of product or equipment.
1. Mark up each submittal to show which products and options are applicable.
2. Include the following information, as applicable:
a. Manufacturer's written recommendations.
b. Manufacturer's product specifications.
c. Manufacturer's installation instructions.
d. Manufacturer's catalog cuts.
e. Wiring diagrams showing factory-installed wiring.
f. Compliance with recognized trade association standards.
g. Compliance with recognized testing agency standards.

C. Required Action Submittals
1. Product Data: See Technical Specification Sections

3.4 INFORMATIONAL SUBMITTALS

A. Prepare and submit Informational Submittals required in the Technical Specifications in this contract, if applicable, and as indicated below:

B. Required Informational Submittals:
   1. Accident Prevention Plan.
   2. Construction Schedule.
   3. Current licensure and/or certifications for individuals indicated in QUALIFICATIONS paragraph above or as required in individual technical sections.

3.5 PROGRESS REPORTS

A. The contractor shall prepare and submit weekly progress reports during the course of construction. The reports shall include a narrative description of work in progress and completed during the week with a note on any delays not accounted for in the approved progress schedule. Photos of the work in place shall also be included; photos should consist of both macro (close up detail –e.g. to demonstrate finish quality or equipment data plate) and overview photos that demonstrate the construction phase, larger areas of work in the context of the surrounding area. Photos shall document key construction areas e.g. primary structural member installation, utilities that will be covered by later construction, any damage discovered or caused by the construction and shall be annotated with location information. A substantial completion report shall also be submitted documenting fulfillment of contract requirements. If a punchlist is created; the Contractor shall provide a punchlist completion report prior to contract closeout.

3.6 CLOSEOUT SUBMITTALS

A. Prepare and submit Closeout Submittals, as follows:
   1. See Technical Specification Sections.
   2. One-Year Warranty on Entire Project as required by FAR 52.246–21
   3. Project Record Documents.

PART 4 – TEMPORARY FACILITIES AND CONTROLS

4.1 USE OF UTILITIES

A. Utilities: Use of Government existing facilities will be permitted without metering and without payment of use charges, as long as facilities are cleaned and maintained in an acceptable condition. At Substantial Completion, restore these facilities to condition existing before initial use. The following are available for use by the contractor: water, and electrical power services. The following are not available: telephone service. The contractor is expected to provide whatever phone service he deems necessary for the duration of the project.

4.2 SANITARY FACILITIES

A. Sanitary facilities are available for use by the Contractor. Facilities must be cleaned and maintained daily.

4.3 STAGING AND STORAGE

A. Coordinate on-site staging and storage requirements with the Center Point of Contact or the Contracting Officer Representative (COR).

4.4 CONSTRUCTION WASTE DISPOSAL FACILITIES

A. Provide waste-collection containers in sizes adequate to handle waste from construction operations. Containerize and clearly label hazardous, dangerous, or unsanitary waste materials separately from other waste. Use of Government waste containers will not be permitted.

4.5 SAFETY AND ENVIRONMENTAL PROTECTION

A. Environmental Protection: Contractor shall provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and that minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.

B. Due to the age of most Job Corps structures, any painted surfaces should be considered potentially lead painted; these facilities are not considered child occupied facilities subject to the RRP but contractors disturbing painted surfaces via cutting, sanding, grinding, and similar dust generating activities shall adhere to the principals of lead safe work practices including as applicable wetting surfaces before disturbance to minimize dust generation, capture with HEPA vacuum when dust generation can't be avoided, personal protective equipment use for employees, and prompt cleanup and disposal of waste generated by construction and demolitions activities.

C. Barricades, Warning Signs, and Lights: Comply with the national or state Manual on Uniform Traffic Control Devices (MUTCD) for erecting structurally adequate barricades, paint with appropriate colors, graphics, and warning signs/lights to inform personnel, students, and the public of possible hazards.

D. Fire Protection.
   1. Store combustible materials in containers in fire-safe locations.
   2. Maintain unobstructed access to fire extinguishers, fire hydrants, fire-protection facilities, stairways, and other access routes for firefighting. Prohibit smoking in hazardous fire-exposure areas.
   3. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition.

PART 5 – EXECUTION REQUIREMENTS

5.1 EXAMINATION

A. Before beginning work, examine areas and document existing conditions. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

B. Field verify all new and existing dimensions affecting the work of this contract before ordering products.

C. Immediately on discovery of the need for clarification of the Contract Documents, submit a request for information to CO. Include a detailed description of problem encountered, together with recommendations for changing the Contract Documents.

5.2 INSTALLATION

A. Locate the Work and components of the Work accurately, in correct alignment and elevation to fit with existing construction.

B. Make vertical work plumb and make horizontal work level.

C. Install per manufacturer's written instructions and recommendations.

5.3 PROGRESS CLEANING

A. Clean Project site and work areas daily, including common areas. Dispose of materials lawfully.
   1. Comply with NFPA 241 requirements for removal of combustible waste materials and debris.
   2. Do not hold materials more than 7 days during normal weather or 3 days if the temperature is expected to rise above 80 deg F (27 deg C).

B. Burying or burning waste materials on-site will not be permitted. Washing waste materials down sewers or into waterways will not be permitted.

5.4 PROTECTION OF INSTALLED CONSTRUCTION

A. Provide protection and maintain conditions that ensure installed Work does not get damaged or deteriorated prior to Substantial Completion.

5.5 CORRECTION OF THE WORK

A. Repair or remove and replace defective construction. Restore damaged substrates and finishes.
   1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.

PART 6 – CLOSEOUT PROCEDURES

6.1 SUBSTANTIAL COMPLETION

A. Definition of Substantial Completion: The Date certified by the CO when construction is sufficiently complete, in accordance with the Contract Documents, so the Government can occupy or utilize the Work or designated portion thereof for the use for which it is intended, as expressed in the Contract Documents.

B. Preliminary Procedures: Before requesting inspection for determining date of Substantial Completion, complete the following. List items below that are incomplete in request.
   1. Prepare a list of items to be completed and corrected (punch list), the value of items on the list, and reasons why the Work is not complete.
   2. Advise the CO of pending insurance changeover requirements.
   3. Submit specific warranties, workmanship bonds, final certifications, and similar documents.
   4. Obtain and submit releases permitting Government unrestricted use of the Work and access to services and utilities.
   5. Deliver tools, spare parts, extra materials, and similar items to location designated by Government. Label with manufacturer's name and model number where applicable.
   6. Complete startup testing of systems.
   7. Submit changeover information related to Government occupancy, use, operation, and maintenance.
   8. Complete final cleaning requirements, including touchup painting.
   9. Touch up and otherwise repair and restore marred exposed finishes to eliminate visual defects.

C. Inspection: Submit a written request for inspection for Substantial Completion. On receipt of request, CO will either proceed with inspection or notify Contractor of unfulfilled requirements. CO will prepare the Certificate of Substantial Completion after inspection or will notify Contractor of items, either on Contractor's list or additional items identified by CO, that must be completed or corrected before certificate will be issued.
   1. Reinspection: Request reinspection when the Work identified in previous inspections as incomplete is completed or corrected.
   2. Results of completed inspection will form the basis of requirements for Final Completion.

6.2 FINAL COMPLETION

A. Preliminary Procedures: Before requesting final inspection for determining date of Final Completion, complete the following:

1. Submit certified copy of CO's Substantial Completion inspection list of items to be completed or corrected (punch list), endorsed and dated by CO. The certified copy of the list shall state that each item has been completed or otherwise resolved for acceptance.
2. Submit evidence of final, continuing insurance coverage complying with insurance requirements.
3. Instruct Government personnel in operation, adjustment, and maintenance of products, equipment, and systems.

B. Inspections: Submit a written request for final inspection for acceptance. On receipt of request, CO will either proceed with inspection or notify Contractor of unfulfilled requirements. CO will prepare a final Certificate for Payment after inspection or will notify Contractor of construction that must be completed or corrected before certificate will be issued.
   1. Reinspection: Request reinspection when the Work identified in previous inspections as incomplete is completed or corrected.

6.3 LIST OF INCOMPLETE ITEMS (PUNCH LIST)

A. Preparation: Submit two copies of list. Include name and identification of each space and area affected by construction operations for incomplete items and items needing correction including, if necessary, areas disturbed by Contractor that are outside the limits of construction.

6.4 PROJECT RECORD DOCUMENTS

A. General: Do not use Project Record Documents for construction purposes. Protect Project Record Documents from deterioration and loss. Provide access to Project Record Documents for CO's reference during normal working hours.

B. Provide Draft of Project Record Drawings at Substantial Completion. CO will return reviewed copy. Provide Final Project Record Drawings at Final Completion.

C. Record Drawings: Maintain and submit one set of blue- or black-line white prints of Contract Drawings and Shop Drawings.

   1. Mark Record Prints to show the actual installation where installation varies from that shown originally. Require individual or entity who obtained record data, whether individual or entity is Installer, subcontractor, or similar entity, to prepare the marked-up Record Prints.

      a. Give particular attention to information on concealed elements that cannot be readily identified and recorded later.
      b. Record data as soon as possible after obtaining it. Record and check the markup before enclosing concealed installations.

2. Mark record sets with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of the Work at the same location.
3. Note Construction Change Directive numbers, Change Order numbers, alternate numbers, and similar identification where applicable.
4. Identify and date each Record Drawing; include the designation "PROJECT RECORD DRAWING" in a prominent location. Organize into manageable sets; bind each set with durable paper cover sheets. Include identification on cover sheets.

6.5 FINAL CLEANING

A. General: Provide final cleaning. Conduct cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.

B. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.

C. Complete the following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:
1. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
2. Remove tools, construction equipment, machinery, and surplus material from Project site.
3. Touch up and otherwise repair and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.

D. Remove waste materials from Project site and dispose of lawfully.

END OF SECTION 010150

SECTION 017329 – CUTTING AND PATCHING

PART 1 - GENERAL

1.1 RESPONSIBILITIES

A. If the Project involves a General Construction Contractor, the General Construction Contractor shall be responsible for the cutting of openings with area equal to or greater than thirty-six (36) square inches. The cutting of openings with area less than thirty-six (36) square inches shall be the responsibility of the Separate Prime Contractor whose work requires the cutting. The General Construction Contractor shall be responsible for all patching work, regardless of the size of the opening.

B. If the cutting of openings into newly constructed walls, partitions, floors, or ceilings is required due to lack of coordination between Separate Prime Contractors, or due to ill-timed or defective work by one or more Separate Prime Contractors, the Separate Prime Contractor primarily responsible for the lack of coordination or the ill-timed or defective work shall bear all costs involved in cutting and patching. The Government may direct the General Construction Contractor to actually perform the cutting and patching.

C. If the Project does not involve a General Construction Contractor, cutting and patching shall be the responsibility of the Separate Prime Contractor whose work requires the cutting. The Contractor shall employ competent individuals, approved by the Government, to do the cutting and patching.

1.2 SUBMITTALS

A. When cutting and patching procedures are required to be approved prior to the work, the Contractor shall submit a plan for cutting and patching well in advance of the work. The submittal shall describe the procedures and request approval to proceed. The submittal shall include the following, as applicable and be submitted per 010150 Submittal procedures.
   1. A description of the limits of cutting and patching required, how it is to be performed, and why it cannot be avoided.
   2. A description of the anticipated results in terms of changes to existing construction, including changes to structural elements and operating components as well as changes in the building's appearance and other significant visual elements.
   3. A listing of products to be used and firms or entities that will perform work.
   4. Identification of the dates when cutting and patching is to be performed.
   5. Identification of any utilities that will be distributed or affected, including those that will be relocated and those that will be temporarily out of service, and an indication of how long service will be disrupted.
   6. Where cutting and patching involves addition of reinforcement to structural elements, details and engineering calculations to show how reinforcement is integrated with the original structure.

B. Approval by the Government to proceed with cutting and patching does not waive the right of the Government to later require complete removal and replacement of any part of the work found to be unsatisfactory.

1.3 MATERIALS

A. Materials used shall be identical to existing materials.

B. If identical materials are not available or cannot be used where exposed surfaces are involved, materials used shall match existing adjacent surfaces to the fullest extent possible with regard to visual effect.

C. All materials used shall have installed performance equal or greater than that of existing materials.

D. All materials used shall be compatible with adjacent existing materials and shall have no deleterious effect on the existing construction.

1.4 INSPECTION

A. Before cutting existing surfaces, the Contractor shall examine surfaces to be cut and patched and conditions under which cutting and patching are to be performed.

B. The Contractor shall avoid any unsafe or unsatisfactory conditions, and if unsafe or unsatisfactory conditions are encountered, corrective action shall be taken before proceeding.

1.5 PREPARATION

A. Work to be cut shall be provided with temporary support.

B. Existing construction shall be protected during cutting and patching to prevent damage by cutting and patching operations.

C. Protection from adverse weather conditions shall be provided for portions of the project that might be exposed during cutting and patching operations.

D. Cutting and patching operations shall be conducted so as to avoid interference with use of adjoining areas or interruption of free passage to adjoining areas.

E. The Contractor shall take all precautions necessary to avoid cutting existing pipe, conduit, or ductwork that serve the facility and are scheduled to be removed or relocated, until provisions have been made for bypass installations.

1.6 PERFORMANCE, GENERAL

A. Skilled workmen shall be employed to perform cutting and patching.

B. Cutting and patching shall be undertaken at the earliest feasible time and completed without delay.

C. Cutting, fitting, and patching for installation of components or the performance of other construction activities shall be performed so as to restore surfaces to their original condition.

D. Openings shall be temporarily covered when not in use.

E. Where excavating and backfilling are required, the Contractor shall comply with requirements of applicable sections of the Technical Specifications.

F. For utility services shown or required to be removed, relocated, or abandoned, the Contractor shall ensure the utility service is shut off or not active prior to performing any work on the system. If the utility service requires continued operation during the Work, the Contractor shall by-pass the service before cutting service pipe or conduit. Any pipe or conduit that is to remain shall be sealed with a cap, valve, or plug to prevent entrance of moisture or other foreign matter.

1.7 PERFORMANCE, CUTTING

A. Cutting of existing construction shall be performed using methods least likely to damage elements to be retained or adjoining construction. Where possible, proposed procedures shall be reviewed with the original installer and shall comply with the original installer's recommendations.

B. Where cutting is required, hand or small power tools designed for sawing or grinding shall be used. Hammering and chopping shall not be done. Holes and slots shall be cut neatly to the size required with minimum disturbance of adjacent surfaces.

C. To avoid marring existing finished surfaces, cutting or drilling shall be done from the exposed or finished side into concealed surfaces.

D. A cutting machine such as a carborundum saw, or diamond core drill shall be used to cut through concrete and masonry.

1.8 PERFORMANCE, PATCHING

A. Patching shall be performed with durable seams that are as invisible as possible. Patching shall meet specified tolerances.

B. Where feasible, patched areas shall be inspected and tested to demonstrate integrity of the installation.

C. Exposed finishes of patched areas shall be restored, and finished restoration shall be extended into retained adjoining construction in a manner that will eliminate evidence of patching and refinishing.

D. Where patching occurs in a smooth painted surface, the final paint coat shall be extended over entire wall plane (wall corner to corner) containing the patch. The patched area itself shall receive a primer and second coat.

E. Where the removal of walls or partitions extends from the new space into another existing space, the floor and wall surfaces in the new space shall be patched and repaired to provide an even surface of uniform color and appearance. If uniform color and appearance cannot be achieved, the existing floor and wall coverings in the existing space shall be removed and replaced with new materials.

F. Existing ceilings shall be patched, repaired, or re-hung as necessary to provide an even plane surface of uniform appearance.

1.9 COORDINATION

A. Structural elements shall not receive any cutting and patching that would reduce their load-carrying capacity or load-deflection ratio. Structural elements include:
   1. Foundation construction
   2. Bearing and retaining walls
   3. Structural concrete
   4. Structural steel
   5. Lintels
   6. Structural decking
   7. Stair systems
   8. Miscellaneous structural metals
   9. Exterior curtain wall construction
   10. Equipment supports.
   11. Piping, ductwork, vessels, and equipment
   12. Structural systems of Special Construction in the Technical Specifications

B. Safety-related components and operating elements shall not receive cutting and patching in a manner that would result in reducing their capacity to perform as intended, result in increased maintenance, or result in decreased operational life or safety. For those components and elements covered by a warranty, the Contractor shall coordinate cutting and patching with the warranting party, and the cutting and patching shall not void the warranty. Safety-related components and operating elements include:
   1. Shoring, bracing, and sheeting.
   2. Primary operational systems and equipment
   3. Air or smoke barriers
   4. Water, moisture, or vapor barriers
   5. Membranes and flashings
   6. Roofing systems and membranes
   7. Fire protection systems
   8. Noise and vibration control elements and systems
   9. Control systems
   10. Communication systems
   11. Conveying systems
   12. Electrical wiring systems
   13. Special Construction specified in the Technical Specifications

C. Visually-exposed components and construction that is exposed on the exterior or exposed in occupied spaces shall not receive cutting and patching in a manner that will reduce the building's aesthetic qualities or result in visual evidence of cutting and patching. If possible, the original installer or fabricator shall be retained to cut and patch the following categories of exposed work:
   1. Processed concrete finishes.
   2. Ornamental metal
   3. Matched-veneer woodwork.
   4. Window wall system
   5. Stucco and ornamental plaster
   6. Acoustical ceilings
   7. Carpeting
   8. HVAC enclosures, cabinets, or covers.
   9. If it is not possible to engage the original installer or fabricator, another recognized, experienced, and specialized firm shall be engaged.
   10. For those components and construction covered by a warranty, the Contractor shall coordinate cutting and patching with the warranting party, and the cutting and patching shall not void the warranty.

D. The General Construction Contractor shall be responsible for removing, storing, and replacing removable ceilings where work is to be provided above an existing removable ceiling not indicated for replacement.

1.10 CLEANING AND RESTORATION

A. Thorough cleaning shall be performed for areas and spaces where cutting and patching is performed or used as access.

B. Paint, mortar, oils, putty, and items of similar nature shall be completely removed.

C. Piping, conduit, and similar features shall be thoroughly cleaned before painting or other finishing is applied.

D. Pipe covering and insulation damaged incidental to cutting and patching shall be restored to its original condition.

PART 2 - PRODUCTS

2.1 (NOT USED)

PART 3 - EXECUTION

3.1 (NOT USED)

END OF SECTION 017329

SECTION 024119 - SELECTIVE DEMOLITION

PART 1 - GENERAL

1.1 SUMMARY

A. The Work of this Section Includes:

1. Demolition and removal of selected portions of exterior or interior of building or structure and site elements.
2. Removal of existing items for reinstallation.

B. Related Requirements:

1. Section 010150 "General Requirements" for restrictions on use of the premises, Government-occupancy requirements, and phasing requirements.
2. Section 017329 "Cutting and Patching" for cutting and patching procedures.

1.2 MATERIALS OWNERSHIP

A. Unless otherwise indicated, demolition waste becomes property of Contractor.

1.3 INFORMATIONAL SUBMITTALS

A. Survey of Existing Conditions: Submit survey.

B. Proposed Protection Measures: Submit report, including Drawings, that indicates the measures proposed for protecting individuals and property, for environmental protection, for dust control and, for noise control. Indicate proposed locations and construction of barriers.

C. Schedule of selective demolition activities with starting and ending dates for each activity.

1.4 FIELD CONDITIONS

A. Government will occupy portions of building immediately adjacent to selective demolition area. Conduct selective demolition so Government's operations will not be disrupted.

B. Conditions existing at time of inspection for bidding purpose will be maintained by Government as far as practical.

C. Notify Contracting Officer of discrepancies between existing conditions and Drawings before proceeding with selective demolition.

D. Hazardous Materials:

1. It is not expected that hazardous materials will be encountered in the Work.

a. If materials suspected of containing hazardous materials are encountered, do not disturb; immediately notify Contracting Officer.

E. On-site sale of removed items or materials is not permitted.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Regulatory Requirements: Comply with governing EPA notification regulations before beginning selective demolition. Comply with hauling and disposal regulations of authorities having jurisdiction.

B. Standards: Comply with ANSI/ASSP A10.6 and NFPA 241.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify that utilities have been disconnected and capped before starting selective demolition operations.

B. Survey of Existing Conditions: Record existing conditions by use of preconstruction photographs or video.

1. Inventory and record the condition of items to be removed for salvage or reinstallation. Photograph or video conditions that might be misconstrued as damage caused by removal.
2. Photograph or video existing conditions of adjoining construction including finish surfaces, that might be misconstrued as damage caused by selective demolition operations or removal of items for salvage or reinstallation.

3.2 PREPARATION

A. Temporary Shoring: Design, provide, and maintain shoring, bracing, and structural supports as required to preserve stability and prevent movement, settlement, or collapse of construction and finishes to remain, and to prevent unexpected or uncontrolled movement or collapse of construction being demolished.

B. Temporary Protection: Provide temporary barricades and other protection required to prevent injury to people and damage to adjacent buildings and facilities to remain.

C. Existing Items to Remain: Protect construction indicated to remain against damage and soiling during selective demolition. When permitted by Contracting Officer, items may be removed to a suitable, protected storage location and reinstalled in their original locations after selective demolition operations are complete.

3.3 UTILITY SERVICES AND BUILDING SYSTEMS

A. Existing Services/Systems to Remain: Maintain utilities and building systems and equipment to remain and protect against damage during selective demolition operations.

B. Existing Services/Systems to Be Removed, Relocated, or Abandoned: Locate, identify, disconnect, and seal or cap off utilities and building systems serving areas to be selectively demolished.
   1. Arrange to shut off utilities with utility companies.
   2. If disconnection of utilities and building systems will affect adjacent occupied parts of the building, provide temporary services/systems that bypass area of selective demolition and that maintain continuity of services/systems to those parts of the building.
   3. Demolish and remove existing building systems, equipment, and components indicated on Drawings to be removed.

      a. Piping to Be Removed: Remove portion of piping indicated to be removed and cap or plug remaining piping with same or compatible piping material.
      b. Ducts to Be Removed: Remove portion of ducts indicated to be removed and plug remaining ducts with same or compatible ductwork material.
      c. Equipment to Be Removed: Disconnect and cap services and remove equipment and components.

   4. Abandon existing building systems, equipment, and components indicated on Drawings to be abandoned in place.

      a. Piping to Be Abandoned in Place: Drain piping and cap or plug piping with same or compatible piping material and leave in place.
      b. Ducts to Be Abandoned in Place: Cap or plug ducts with same or compatible ductwork material and leave in place.

   5. Remove and reinstall/salvage existing building systems, equipment, and components indicated on drawings to be removed and reinstalled or removed and salvaged:

      a. Equipment to Be Removed and Reinstalled: Disconnect and cap services and remove, clean, and store equipment and components; when appropriate, reinstall, reconnect, and make equipment operational.
      b. Equipment to Be Removed and Salvaged: Disconnect and cap services and remove equipment and components and deliver to Government’s designated representative.

3.4 SALVAGE/REINSTALL

A. Removed and Reinstalled Items:

   1. Clean and repair items to functional condition adequate for intended reuse.
   2. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use indicated.

3.5 SELECTIVE DEMOLITION, GENERAL

A. General: Demolish and remove existing construction only to extent required by new construction and as indicated. Use methods required to complete the Work within limitations of governing regulations and as follows:

1. Neatly cut openings and holes plumb, square, and true to dimensions required. Use cutting methods least likely to damage construction to remain or adjoining construction. Use hand tools or small power tools designed for sawing or grinding, not hammering and chopping. Temporarily cover openings to remain.
2. Cut or drill from the exposed or finished side into concealed surfaces to avoid marring existing finished surfaces.
3. Do not use cutting torches until work area is cleared of flammable materials. At concealed spaces, such as duct and pipe interiors, verify condition and contents of hidden space before starting flame-cutting operations. Maintain portable fire-suppression devices during flame-cutting operations.
4. Maintain fire watch during and for at least 1 hours after flame-cutting operations.
5. Locate selective demolition equipment and remove debris and materials so as not to impose excessive loads on supporting walls, floors, or framing.

B. Site Access and Temporary Controls: Conduct selective demolition and debris-removal operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

3.6 DISPOSAL OF DEMOLISHED MATERIALS

A. Remove demolition waste materials from Project site and dispose of them in an EPA-approved construction and demolition waste landfill acceptable to authorities having jurisdiction.

1. Do not allow demolished materials to accumulate on-site.
2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.

B. Burning: Do not burn demolished materials.

3.7 CLEANING

A. Clean adjacent structures and improvements of dust, dirt, and debris caused by selective demolition operations. Return adjacent areas to condition existing before selective demolition operations began.

END OF SECTION 024119

SECTION 061000 - ROUGH CARPENTRY

PART 1 - GENERAL

1.1 SECTION REQUIREMENTS

A. Submittals: ICC-ES evaluation reports for wood-preservative treated wood, engineered wood products, and metal framing anchors.

PART 2 - PRODUCTS

2.1 WOOD PRODUCTS, GENERAL

A. Lumber: Provide dressed lumber, S4S, marked with grade stamp of inspection agency.

2.2 TREATED MATERIALS

A. Preservative-Treated Materials: AWPA U1; Use Category UC3b for wood framing within 18" of the ground and Use Category UC4a for items in contact with the ground or exterior concrete or masonry.

1. Use treatment containing no arsenic or chromium. Do not use inorganic boron (SBX) for sill plates.
2. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent.
3. Mark lumber with treatment quality mark of an inspection agency approved by the ALSC Board of Review.

2.3 FRAMING, FASCIA, PURLINS, AND FURRING

A. Dimension Lumber:

1. Maximum Moisture Content: 15 percent for 2-inch nominal thickness or less, 19 percent for more than 2-inch nominal thickness.

2. Lumber:
   a. No. 2 or better of any species.

2.4 MISCELLANEOUS PRODUCTS

A. Fasteners: Size and type indicated. Where rough carpentry is exposed to weather, in ground contact, or in area of high relative humidity, provide fasteners with hot-dip zinc coating complying with ASTM A 153/A 153M or of Type 304 stainless steel.

1. Power-Driven Fasteners: CABO NER-272.
2. Bolts: Steel bolts complying with ASTM A 307, Grade A; with ASTM A 563 hex nuts and, where indicated, flat washers.

3. Structural Screw: GRK R4 or equal: Coated or Stainless Steel, Evaluated for compliance with IBC for wood structural fastening.
    a. Size based on pull test values and roofing system manufacturer installation instructions and shop drawings.

B. Flexible Flashing: Self-adhesive product consisting of a butyl rubber or rubberized-asphalt compound, bonded to a backing sheet to produce an overall thickness of not less than 0.025 inch.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Set rough carpentry to required levels and lines, with members plumb, true to line, cut, and fitted. Locate nailers, blocking, and similar supports to comply with requirements for attaching other construction.

B. Framing Standard: Comply with AF&PA's WCD 1, "Details for Conventional Wood Frame Construction," unless otherwise indicated.

C. Do not splice structural members between supports unless otherwise indicated.

D. Securely attach rough carpentry to substrates, complying with the following:

1. CABO NER-272 for power-driven fasteners.
2. Published requirements of metal framing anchor manufacturer.
3. Table 2304.9.2, "Fastening Schedule," in the IBC.

END OF SECTION 061000

SECTION 075419 - POLYVINYL-CHLORIDE (PVC) ROOFING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Mechanically fastened (metal retrofit) polyvinyl chloride (PVC) roofing system.
2. Accessory roofing materials.
3. Roof insulation.
4. Insulation accessories.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: Include roof plans, sections, details, and attachments to other work, including the following:

1. Layout and thickness of insulation.
2. Base flashings and membrane terminations.
3. Flashing details at penetrations.
4. Roof plan showing orientation of steel roof deck and orientation of roof membrane, fastening spacings, and patterns for mechanically fastened roofing system.
5. Insulation fastening patterns for corner, perimeter, and field-of-roof locations.

C. Wind Uplift Resistance Submittal: For roofing system, indicating compliance with wind uplift performance requirements. Basic wind speed is indicated on the drawings. Manufacturer’s engineering services shall be engaged for provision of fastener requirements based upon International Building Code and/or ASCE 7-22 design specifications.

1.3 INFORMATIONAL SUBMITTALS

A. Manufacturer Certificates:

1. Performance Requirement Certificate: Signed by roof membrane manufacturer, certifying that roofing system complies with requirements specified in "Performance Requirements" Article.

a. Submit evidence of compliance with performance requirements.

2. Special Warranty Certificate: Signed by roof membrane manufacturer, certifying that all materials supplied under this Section are acceptable for special warranty.

B. Product Test Reports: For roof membrane and insulation, tests performed by independent qualified testing agency indicating compliance with specified requirements.

C. Research reports.

D. Field Test Reports:
   1. Fastener-pullout test results and manufacturer's revised requirements for fastener patterns.

E. Field quality-control reports.

F. Sample warranties.

1.4 CLOSEOUT SUBMITTALS

A. Maintenance data.

B. Certified statement from existing roof membrane manufacturer stating that existing roof warranty has not been affected by Work performed under this Section.

1.5 QUALITY ASSURANCE

A. Qualifications:

   1. Manufacturers: A qualified manufacturer that is UL listed for roofing system identical to that used for this Project.
   2. Installers: A qualified firm that is approved, authorized, or licensed by roofing system manufacturer to install manufacturer's product and that is eligible to receive manufacturer's special warranty.

1.6 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of roofing system that fail in materials or workmanship within specified warranty period.

   1. Warranty Period: 20-years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Accelerated Weathering: Roof membrane to withstand 2000 hours of exposure when tested according to ASTM G152, ASTM G154, or ASTM G155.

B. Impact Resistance: Roof membrane to resist impact damage when tested according to ASTM D3746, ASTM D4272/D4272M, or the "Resistance to Foot Traffic Test" in FM Approvals 4470.

C. Material Compatibility: Roofing materials to be compatible with one another and adjacent materials under conditions of service and application required, as demonstrated by roof membrane manufacturer based on testing and field experience.

D. Wind Uplift Resistance: Design roofing system to resist the following wind uplift pressures when tested according to FM Approvals 4474, UL 580, or UL 1897:

   a. Zone area pressures indicated on plans are for initial scoping only. Contractor shall verify field conditions (building heights, roof slope, roof dimensions, etc) and provide information to the roofing system's engineering department for zone pressure calculations. This information along with pullout tests performed by the contractor or manufacturer representative shall be documented on the shop drawings and fastening schedule.

E. Energy Star Listing: Roofing system to be listed on the DOE's Energy Star "Roof Products Qualified Product List" for roof products.

F. Exterior Fire-Test Exposure: ASTM E108 or UL 790, Class A, Class B, or Class C; for application and roof slopes indicated; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

G. Fire-Resistance Ratings: Comply with fire-resistance-rated assembly designs indicated. Identify products with appropriate markings of applicable testing agency.
   1. Basis of Design System is Class C, UL listed for the intended application, see UL Product IQ/Durolast TGFU.R10128.
      a. Class C, Mechanically Fastened (Class B with incline restriction)
         1) Deck: Non-Combustible (NC).
         2) Incline: Unlimited. (3:12 for Class B).
         3) Insulation: Duro-guard ISO-II-H. (flute fill + 2 layers 1.5" continuous).
         4) Membrane: Duro-Tuff (PVC) 60-mil.

2.2 POLYVINYL CHLORIDE (PVC) ROOFING

A. PVC Sheet Type III: ASTM D4434/D4434M, fabric reinforced: weft inserted 18x9 polyester fabric construction with PVC film formulation of resins, plasticizers, biocides, flame retardents, and U.V. absorbents.

   1. Duro-last, Duro-Tuff, 60-mil (DT-60) Membrane.
   2. Thickness: Minimum 60 mils.
   3. Exposed Face Color: White.
   4. Solar reflectance:
      a. Initial: 0.86.
      b. 3-year: 0.74
   5. Thermal Emittance:
      a. Initial: 0.87

b. 3-year: 0.87.

6. Solar Reflectance:
   a. Initial: 108.
   b. 3-year: 91.

2.3 ACCESSORY ROOFING MATERIALS

A. General: Accessory materials recommended by roofing system manufacturer for intended use and compatible with other roofing components.

1. Adhesives and Sealants: Comply with VOC limits of authorities having jurisdiction.

B. Sheet Flashing: Manufacturer's standard sheet flashing of same material, type, reinforcement, thickness, and color as PVC sheet.

C. Prefabricated Pipe Flashings: As recommended by roof membrane manufacturer.

D. Bonding Adhesive: Manufacturer's standard.

E. Water-Based, Fabric-Backed Membrane Adhesive: Roofing system manufacturer's standard water-based, cold-applied adhesive formulated for compatibility and use with fabric-backed membrane roofing.

F. Low-Rise, Urethane, Fabric-Backed Membrane Adhesive: Roof system manufacturer's standard spray-applied, low-rise, two-component urethane adhesive formulated for compatibility and use with fabric-backed membrane roofing.

G. Metal Termination Bars: Manufacturer's standard, predrilled stainless steel or aluminum bars, approximately 1 by 1/8 inch thick; with anchors.

H. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Approvals 4470, designed for fastening roofing components to substrate, and acceptable to roofing system manufacturer.

I. Miscellaneous Accessories: Provide pourable sealers, preformed cone and vent sheet flashings, preformed inside and outside corner sheet flashings, T-joint covers, lap sealants, termination reglets, and other accessories.

2.4 ROOF INSULATION

A. Polyisocyanurate Board Insulation: ASTM C1289, Type II, Class 1, Grade 3, closed-cell, foam core insulation board with integrally laminated, fiber-reinforced facer compatible with roof membrane.

1. Basis of Design Product: Duro-Guard ISO II-H.
2. Size: 48 by 48 inches or 48 by 96 inches as required/compatible with manufacturer’s installation instructions.
3. Thickness:

a. Flute Fill: Contractor to verify existing metal roof profiles and provide flute fill dimensions that closely match existing metal profiles and comply with the flute span rating of insulation base layer above.

b. Base Layer: 1-1/2 inches.
c. Upper Layer: 1-1/2 inches.

2.5 INSULATION ACCESSORIES

A. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Approvals 4470, designed for fastening roof insulation to substrate, and acceptable to roofing system manufacturer.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

1. Verify that surface plane flatness and fastening of the existing roof metal complies with requirements of the manufacturer’s requirements contained in Durolast Mechanically Fastened System specification (retrofit).

3.2 PREPARATION

A. Perform fastener-pullout tests according to roof system manufacturer's written instructions.

1. Submit test result within 24 hours of performing tests.

a. Include manufacturer's requirements for any revision to previously submitted fastener patterns required to achieve specified wind uplift requirements.

3.3 INSTALLATION OF ROOFING, GENERAL

A. Install roofing system according to roofing system manufacturer's written instructions and UL listed roof assembly requirements.

B. Complete terminations and base flashings and provide temporary seals to prevent water from entering completed sections of roofing system at end of workday or when rain is forecast. Remove and discard temporary seals before beginning work on adjoining roofing.

C. Install roof membrane and auxiliary materials to tie into existing roofing to maintain weathertightness of transition.

3.4 INSTALLATION OF INSULATION

A. Coordinate installing roofing system components so insulation is not exposed to precipitation or left exposed at end of workday.

B. Comply with roofing system and insulation manufacturer's written instructions for installing roof insulation.

C. Installation Over Metal Roofing:

1. Install flute fill insulation, cut to closely fit metal roof panel to comply with base layer span rating, attach in accordance with roof manufacturer's installation instructions.

2. Install base layer of insulation with joints staggered not less than 12 inches in adjacent rowsand with long joints continuous at right angle to flutes of decking.

a. Locate end joints over crests of decking unless otherwise allowed by roofing system installation instructions.
b. Trim insulation neatly to fit around penetrations and projections, and to fit tight to intersecting sloping roof decks.
c. Make joints between adjacent insulation boards not more than 1/4 inch in width.
d. Fill gaps exceeding 1/4 inch with insulation.
e. Cut and fit insulation within 1/4 inch of nailers, projections, and penetrations.
f. Mechanically attach base layer of insulationusing mechanical fasteners specifically designed and sized for fastening specified board-type roof insulation to metal decks.
1) Fasten insulation to resist specified uplift pressure at corners, perimeter, and field of roof.

3. Install upper layers of insulation with joints of each layer offset not less than 6 inches from previous layer of insulation.

a. Staggered end joints within each layer not less than 24 inches in adjacent rows.
b. Install with long joints continuous and with end joints staggered not less than 12 inches in adjacent rows.
c. Trim insulation neatly to fit around penetrations and projections, and to fit tight to intersecting sloping roof decks.
d. Make joints between adjacent insulation boards not more than 1/4 inch in width.
e. Trim insulation so that water flow is unrestricted.
f. Fill gaps exceeding 1/4 inch with insulation.
g. Cut and fit insulation within 1/4 inch of nailers, projections, and penetrations.

3.5 INSTALLATION OF MECHANICALLY FASTENED ROOF MEMBRANE

A. Mechanically fasten roof membrane over area to receive roofing according to roofing system manufacturer's written instructions.

B. Unroll roof membrane and allow to relax before installing.

C. For in-splice attachment, install roof membrane with long dimension perpendicular to steel roof deck flutes.

D. Start installation of roofing in presence of roofing system manufacturer's technical personnel.

E. Accurately align roof membrane, and maintain uniform side and end laps of minimum dimensions required by manufacturer. Stagger end laps.

F. Mechanically fasten or adhere roof membrane securely at terminations, penetrations, and perimeter of roofing.

G. Apply roof membrane with side laps shingled with slope of roof deck where possible.

H. In-Seam Attachment: Secure one edge of PVC sheet using fastening plates or metal battens centered within seam, and mechanically fasten PVC sheet to roof deck.

I. Seams: Clean seam areas, overlap roof membrane, and hot-air weld side and end laps of roof membrane and sheet flashings to ensure a watertight seam installation.

1. Test lap edges with probe to verify seam weld continuity. Apply lap sealant to seal cut edges of roof membrane and sheet flashings.
2. Verify field strength of seams a minimum of twice daily, and repair seam sample areas.
3. Repair tears, voids, and lapped seams in roof membrane that do not comply with requirements.

J. Spread sealant bed over deck-drain flange at roof drains, and securely seal roof membrane in place with clamping ring.

3.6 INSTALLATION OF BASE FLASHING

A. Install sheet flashings and preformed flashing accessories, and adhere to substrates according to roofing system manufacturer's written instructions.

B. Apply bonding adhesive to substrate and underside of sheet flashing at required rate, and allow to partially dry. Do not apply to seam area of flashing.

C. Flash penetrations and field-formed inside and outside corners with cured or uncured sheet flashing.

D. Clean seam areas, overlap, and firmly roll sheet flashings into the adhesive. Hot-air weld side and end laps to ensure a watertight seam installation.

E. Terminate and seal top of sheet flashings and mechanically anchor to substrate through termination bars.

3.7 FIELD QUALITY CONTROL

A. Testing Agency: Contractor shall coordinate and engage a qualified testing agency or manufacturer’s representative to inspect substrate conditions, surface preparation, roof membrane application, flashings, protection, and drainage components, and to furnish reports to COR.

B. Final Roof Inspection: Arrange for roofing system manufacturer's technical personnel to inspect roofing installation on completion, in presence of COR, and to prepare inspection report.

C. Repair or remove and replace components of roofing system where inspections indicate that they do not comply with specified requirements.

3.8 PROTECTING AND CLEANING

A. Protect roofing system from damage and wear during remainder of construction period. When remaining construction does not affect or endanger roofing, inspect roofing system for deterioration and damage, describing its nature and extent in a written report, with copies to Architect and Owner.

B. Correct deficiencies in or remove roofing system that does not comply with requirements, repair substrates, and repair or reinstall roofing system to a condition free of damage and deterioration at time of Substantial Completion and according to warranty requirements.

END OF SECTION 075419

SECTION 076200

SHEET METAL FLASHING AND TRIM

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Formed roof-drainage sheet metal fabrications.
   2. Formed wall sheet metal fabrications.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: For sheet metal flashing and trim.

   1. Include plans, elevations, sections, and attachment details.
   2. Distinguish between shop- and field-assembled work.
   3. Include identification of finish for each item.
   4. Include pattern of seams and details of termination points, expansion joints and expansion-joint covers, direction of expansion, roof-penetration flashing, and connections to adjoining work.

C. Samples: For each exposed product and for each color and texture specified.

1.3 INFORMATIONAL SUBMITTALS

A. Product certificates.

B. Product test reports.

C. Sample warranty.

1.4 CLOSEOUT SUBMITTALS

A. Maintenance data.

B. Warranties.

1.5 QUALITY ASSURANCE

A. Fabricator Qualifications: Employs skilled workers who custom fabricate sheet metal flashing and trim similar to that required for this Project and whose products have a record of successful in-service performance.

1. For copings and roof edge flashings that are SPRI ES-1 tested, shop shall be listed as able to fabricate required details as tested and approved.

1.6 WARRANTY

A. Special Warranty on Finishes: Manufacturer agrees to repair finish or replace sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within specified warranty period.

1. Finish Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. General: Sheet metal flashing and trim assemblies shall withstand wind loads, structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Completed sheet metal flashing and trim shall not rattle, leak, or loosen, and shall remain watertight.

B. Sheet Metal Standard for Flashing and Trim: Comply with NRCA's "The NRCA Roofing Manual" and SMACNA's "Architectural Sheet Metal Manual" requirements for dimensions and profiles shown unless more stringent requirements are indicated.

C. Recycled Content of Steel-Sheet Flashing and Trim: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.

D. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.

1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

2.2 SHEET METALS

A. General: Protect mechanical and other finishes on exposed surfaces from damage by applying strippable, temporary protective film before shipping.

B. Aluminum Sheet: ASTM B 209 (ASTM B 209M), alloy as standard with manufacturer for finish required, with temper as required to suit forming operations and performance required.

1. Exposed Coil-Coated Finish:

a. Two-Coat Fluoropolymer: AAMA 620. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

2. Colors: As selected by CO from Manufacturer’s full range.

C. Stainless-Steel Sheet: ASTM A 240/A 240M, Type 304, dead soft, fully annealed; 2D (dull, cold rolled) finish.

D. Metallic-Coated Steel Sheet: Provide zinc-coated (galvanized) steel sheet according to ASTM A 653/A 653M, G90 (Z275) coating designation; prepainted by coil-coating process to comply with ASTM A 755/A 755M.

1. Exposed Coil-Coated Finish:

a. Two-Coat Fluoropolymer: AAMA 621. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

2. Colors: As selected by CO from Manufacturer’s full range.

2.3 MISCELLANEOUS MATERIALS

A. General: Provide materials and types of fasteners, solder, protective coatings, sealants, and other miscellaneous items as required for complete sheet metal flashing and trim installation and as recommended by manufacturer of primary sheet metal or manufactured item unless otherwise indicated.

B. Fasteners: Wood screws, annular threaded nails, self-tapping screws, self-locking rivets and bolts, and other suitable fasteners designed to withstand design loads and recommended by manufacturer of primary sheet metal or manufactured item.

1. General: Blind fasteners or self-drilling screws, gasketed, with hex-washer head.

a. Exposed Fasteners: Heads matching color of sheet metal using plastic caps or factory-applied coating. Provide metal-backed EPDM or PVC sealing washers under heads of exposed fasteners bearing on weather side of metal.
b. Blind Fasteners: High-strength aluminum or stainless-steel rivets suitable for metal being fastened.
c. Spikes and Ferrules: Same material as gutter; with spike with ferrule matching internal gutter width.

2. Fasteners for Aluminum Sheet: Aluminum or Series 300 stainless steel.
3. Fasteners for Stainless-Steel Sheet: Series 300 stainless steel.
4. Fasteners for Zinc-Coated (Galvanized) Steel Sheet: Series 300 stainless steel or hot-dip galvanized steel according to ASTM A 153/A 153M or ASTM F 2329.

C. Solder:
1. For Stainless Steel: ASTM B 32, Grade Sn60, with acid flux of type recommended by stainless-steel sheet manufacturer.
2. For Zinc-Coated (Galvanized) Steel: ASTM B 32, Grade Sn50, 50 percent tin and 50 percent lead or Grade Sn60, 60 percent tin and 40 percent lead.

D. Sealant Tape: Pressure-sensitive, 100 percent solids, polyisobutylene compound sealant tape with release-paper backing. Provide permanently elastic, nonsag, nontoxic, nonstaining tape 1/2 inch (13 mm) wide and 1/8 inch (3 mm) thick.

E. Elastomeric Sealant: ASTM C 920, elastomeric silicone polymer sealant; of type, grade, class, and use classifications required to seal joints in sheet metal flashing and trim and remain watertight.

2.4 FABRICATION, GENERAL

A. General: Custom fabricate sheet metal flashing and trim to comply with details shown and recommendations in cited sheet metal standard that apply to design, dimensions, geometry, metal thickness, and other characteristics of item required. Fabricate sheet metal flashing and trim in shop to greatest extent possible.

1. Obtain field measurements for accurate fit before shop fabrication.
2. Form sheet metal flashing and trim to fit substrates without excessive oil canning, buckling, and tool marks; true to line, levels, and slopes; and with exposed edges folded back to form hems.
3. Conceal fasteners and expansion provisions where possible. Do not use exposed fasteners on faces exposed to view.

B. Expansion Provisions: Form metal for thermal expansion of exposed flashing and trim.

1. Form expansion joints of intermeshing hooked flanges, not less than 1 inch (25 mm) deep, filled with butyl sealant concealed within joints.
2. Use lapped expansion joints only where indicated on Drawings.

C. Sealant Joints: Where movable, nonexpansion-type joints are required, form metal to provide for proper installation of elastomeric sealant according to cited sheet metal standard.

D. Fabricate cleats and attachment devices from same material as accessory being anchored or from compatible, noncorrosive metal.

E. Fabricate cleats and attachment devices of sizes as recommended by cited sheet metal standard for application, but not less than thickness of metal being secured.

F. Seams: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with elastomeric sealant unless otherwise recommended by sealant manufacturer for intended use. Rivet joints where necessary for strength.

2.5 SLOPED ROOF SHEET METAL FABRICATIONS

A. Apron, Step, Cricket, and Baker Flashing: Prepainted, metallic-coated steel, 0.0217 inch thick.

B. Valley Flashing: Zinc-tin alloy-coated stainless steel, 0.018 inch thick.

C. Drip Edges: Prepainted, metallic-coated steel, 0.0217 inch thick.

D. Eave, Rake, Ridge, and Hip Flashing: Prepainted, metallic-coated steel, 0.0217 inch thick.

E. Base Flashing: Prepainted, metallic-coated steel, 0.0276 inch thick.

F. Counterflashing: Prepainted, metallic-coated steel, 0.0217 inch

G. Flashing Receivers: Prepainted, metallic-coated steel, 0.0217 inch.

H. Roof-Penetration Flashing: Prepainted, metallic-coated steel, 0.0276 inch thick.

2.6 FINISHES

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Protect mechanical and painted finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. General: Anchor sheet metal flashing and trim and other components of the Work securely in place, with provisions for thermal and structural movement. Use fasteners, solder, protective coatings, separators, sealants, and other miscellaneous items as required to complete sheet metal flashing and trim system.

1. Install sheet metal flashing and trim true to line, levels, and slopes. Provide uniform, neat seams with minimum exposure of solder, welds, and sealant.
2. Install sheet metal flashing and trim to fit substrates and to result in watertight performance. Verify shapes and dimensions of surfaces to be covered before fabricating sheet metal.
3. Space cleats not more than 12 inches (300 mm) apart. Attach each cleat with at least two fasteners. Bend tabs over fasteners.
4. Install exposed sheet metal flashing and trim with limited oil canning, and free of buckling and tool marks.
5. Torch cutting of sheet metal flashing and trim is not permitted.

B. Metal Protection: Where dissimilar metals contact each other, or where metal contacts pressure-treated wood or other corrosive substrates, protect against galvanic action or corrosion by painting contact surfaces with bituminous coating or by other permanent separation as recommended by sheet metal manufacturer or cited sheet metal standard.

1. Coat concealed side of uncoated-aluminum sheet metal flashing and trim with bituminous coating where flashing and trim contact wood, ferrous metal, or cementitious construction.
2. Underlayment: Where installing sheet metal flashing and trim directly on cementitious or wood substrates, install underlayment and cover with slip sheet.

C. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim. Space movement joints at maximum of 10 feet (3 m) with no joints within 24 inches (600 mm) of corner or intersection.

1. Form expansion joints of intermeshing hooked flanges, not less than 1 inch (25 mm) deep, filled with sealant concealed within joints.
2. Use lapped expansion joints only where indicated on Drawings.

D. Fasteners: Use fastener sizes that penetrate wood blocking or sheathing not less than 1-1/4 inches (32 mm) for nails and not less than 3/4 inch (19 mm) for wood screws.

E. Conceal fasteners and expansion provisions where possible in exposed work and locate to minimize possibility of leakage. Cover and seal fasteners and anchors as required for a tight installation.

F. Seal joints as required for watertight construction. Prepare joints and apply sealants to comply with requirements in Section 079200 "Joint Sealants."

G. Soldered Joints: Clean surfaces to be soldered, removing oils and foreign matter. Pre-tin edges of sheets with solder to width of 1-1/2 inches (38 mm); however, reduce pre-tinning where pre-tinned surface would show in completed Work.

1. Do not solder metallic-coated steel and aluminum sheet.
2. Do not use torches for soldering.
3. Heat surfaces to receive solder, and flow solder into joint. Fill joint completely. Completely remove flux and spatter from exposed surfaces.
4. Stainless-Steel Soldering: Tin edges of uncoated sheets, using solder for stainless steel and acid flux. Promptly remove acid flux residue from metal after tinning and soldering. Comply with solder manufacturer's recommended methods for cleaning and neutralization.
5. Copper Soldering: Tin edges of uncoated sheets, using solder for copper.

H. Rivets: Rivet joints in uncoated aluminum where necessary for strength.

3.2 ROOF FLASHING INSTALLATION

A. General: Install sheet metal flashing and trim to comply with performance requirements, sheet metal manufacturer's written installation instructions, and cited sheet metal standard. Provide concealed fasteners where possible, and set units true to line, levels, and slopes. Install work with laps, joints, and seams that are permanently watertight and weather resistant.

B. Roof Edge Flashing: Anchor to resist uplift and outward forces according to recommendations in cited sheet metal standard unless otherwise indicated. Interlock bottom edge of roof edge flashing with continuous cleat anchored to substrate.

C. Copings: Anchor to resist uplift and outward forces according to recommendations in cited sheet metal standard unless otherwise indicated.

D. Pipe or Post Counterflashing: Install counterflashing umbrella with close-fitting collar with top edge flared for elastomeric sealant, extending minimum of 4 inches (100 mm) over base flashing. Install stainless-steel draw band and tighten.

E. Counterflashing: Coordinate installation of counterflashing with installation of base flashing. Insert counterflashing in reglets or receivers and fit tightly to base flashing. Extend counterflashing 4 inches (100 mm) over base flashing. Lap counterflashing joints minimum of 4 inches (100 mm).

F. Roof-Penetration Flashing: Coordinate installation of roof-penetration flashing with installation of roofing and other items penetrating roof. Seal with butyl sealant and clamp flashing to pipes that penetrate roof.

3.3 CLEANING AND PROTECTION

A. Clean and neutralize flux materials. Clean off excess solder.

B. Clean off excess sealants.

C. Remove temporary protective coverings and strippable films as sheet metal flashing and trim are installed unless otherwise indicated in manufacturer's written installation instructions.

END OF SECTION

SECTION 077100 - ROOF SPECIALTIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes: Manufactured units for the following applications:
   1. Roof-edge specialties.
   2. Roof-edge drainage systems.
   3. Roof Curbs and Extensions.
   4. Roof Jacks

1.2 ACTION SUBMITTALS

A. Product data.

B. Shop Drawings: For roof specialties.

   1. Plans, expansion-joint locations, keyed details, and attachments to other work. Distinguish between factory pre-manufactured- and field-assembled installation.
   2. Details for expansion and contraction; locations of expansion joints, including direction of expansion and contraction.
   3. Indicate profile and pattern of seams and layout of fasteners, cleats, clips, and other attachments.
   4. Details of termination points and assemblies, including fixed points.
   5. Details of special conditions.

1.3 INFORMATIONAL SUBMITTALS

A. Product certificates.

B. Product test reports.

C. Research reports.

D. Sample warranty.

1.4 CLOSEOUT SUBMITTALS

A. Maintenance data.

1.5 WARRANTY

A. Special Warranty on Painted Finishes: Manufacturer agrees to repair finish or replace roof specialties that show evidence of deterioration of factory-applied finishes within specified warranty period.

1. Fluoropolymer Finish: Deterioration includes, but is not limited to, the following:

a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

2. Finish Warranty Period: **10** years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. General Performance: Roof specialties to withstand exposure to weather and resist thermally induced movement without failure, rattling, leaking, or fastener disengagement due to defective substrated.

B. General Specifications in this sections are subject to correction for compliance with roofing system manufacturer’s requirements.

C. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes to prevent buckling, opening of joints, hole elongation, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Provide clips that resist rotation and avoid shear stress as a result of thermal movements. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

2.2 ROOF-EDGE SPECIALTIES

1. Metallic-Coated Steel Fascia Covers: Zinc-coated (galvanized) or aluminum-zinc alloy-coated steel sheet, nominal 0.034-inch thickness.

a. Surface: Smooth, flat finish.
b. Finish: Two-coat fluoropolymer.
c. Color: As selected by COR from manufacturer's full range.

2. Formed Aluminum Fascia Covers: Aluminum sheet, 0.040 inch thick.

a. Surface: Smooth, flat finish.
b. Finish: Two-coat fluoropolymer
c. Color: As selected by COR from manufacturer's full range.

2.3 ROOF-EDGE DRAINAGE SYSTEMS

A. Gutters: Manufactured in uniform section lengths not exceeding 12 ft., with matching corner units, ends, outlet tubes, and other accessories. Elevate back edge at least 1 inch above front edge. Furnish flat-stock gutter straps, gutter brackets, expansion joints, and expansion-joint covers fabricated from same metal as gutters.
   1. Formed Aluminum Sheet: 0.032 inch thick.
   2. Gutter Profile: Style H in accordance with SMACNA's "Architectural Sheet Metal Manual."
      a. Note: Other similar capacity styles can be submitted for approval.
   3. Corners: Factory mitered and soldered, continuously welded, or mechanically clinched and sealed watertight.
   4. Gutter Supports: Manufacturer’s standard support.

B. Downspouts: Plain rectangular or corrugated rectangular complete with machine-crimped, mitered, or smooth-curve elbows, manufactured from the following exposed metal. Furnish with metal hangers, from same material as downspouts, and anchors. Provide with matching elbows to follow roof edge/soffit/siding/and to direct away from building for at grade discharge.
   1. Formed Aluminum Sheet: 0.032 inch thick.
   2. Size: As indicated on Drawings.

C. Conductor Heads: Manufactured conductor heads, each with flanged back and stiffened top edge, and of dimensions and shape indicated, complete with outlet tube that nests into upper end of downspout.
   1. Formed Aluminum Sheet: 0.032 inch thick.
   2. Size: Contractor shall size and coordinate with building construction and gutter/downspout sizes.

D. Finishes:

   1. Aluminum: Two-coat fluoropolymer.

      a. Color: As selected by COR from manufacturer's full range.

E. Accessories:
   1. Adaptors: Provide adaptors for discharge to subdrain systems (when applicable) – verify existing subdrain conditions and coordinate requirements with new gutter system requirements.
   2. Splash pads: Where downspouts do not discharge to pavements or subdrain systems, provide concrete splash pads at discharge elbow to prevent erosion. Splash pads shall be 4,500 psi minimum precast concrete with a formed channel, sloped away from the building foundation, with the following nominal dimensions: 36” long, 12” wide, 3” tall.

2.4 SHEET METAL MATERIALS

A. Metallic-Coated Steel Sheet: Zinc-coated (galvanized) steel sheet complying with minimum ASTM A653/A653M, G90 coating designation, or aluminum-zinc alloy-coated steel sheet complying with minimum ASTM A792/A792M, Class AZ50 coating designation; structural quality.

1. Exposed Coil-Coated Finish: Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

    a. Two-Coat Fluoropolymer Finish: AAMA 2605. System consisting of primer and fluoropolymer color topcoat containing not less than 70 percent PVDF resin by weight in color coat.

2. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester-backer finish consisting of prime coat and wash coat, with a minimum total dry film thickness of 0.5 mil.

B. Aluminum Sheet: ASTM B209/B209M, manufacturer's standard alloy for finish required, with temper to suit forming operations and performance required.

    a. Two-Coat Fluoropolymer Finish: AAMA 2605. System consisting of primer and fluoropolymer color topcoat containing not less than 70 percent PVDF resin by weight in color coat.

2. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester-backer finish consisting of prime coat and wash coat, with a minimum total dry film thickness of 0.5 mil.

2.5 ROOF CURBS AND EXTENSIONS

A. Internally reinforced roof-curb units capable of supporting superimposed live and dead loads, including equipment loads and other construction indicated on Drawings, bearing continuously on roof structure, and capable of meeting performance requirements; with welded or mechanically fastened and sealed corner joints, and integrally formed deck-mounting flange at perimeter bottom.

B. Contractor shall verify existing equipment to receive new curbs or extension and submit products that are fully compatible and providing all features required for equipment function.

C. Size: Coordinate dimensions with roughing-in information existing equipment to be re-supported.

D. Supported Load Capacity: Coordinate load capacity with field verified equipment information.

E. Material:
1. Aluminum: All-welded, 0.0630” minimum thickness.
2. Stainless steel, 18-gauge minimum thickness.

F. Construction:

1. Curb Profile: Manufacturer's standard compatible with roofing system.
2. Fabricate curbs to minimum height of 12 inches above roofing surface unless otherwise indicated.
3. Top Surface: Level top of curb, with roof slope accommodated by sloping deck-mounting flange or by use of leveler frame.

4. Sloping Roofs: Where roof slope exceeds 1/4 inch per 12 inches, fabricate curb with perimeter curb height tapered to accommodate roof slope so that top surface of perimeter curb is level. Equip unit with water diverter or cricket on side that obstructs water flow.
5. Wind-Restraint Straps and Base Flange Attachment: Provide wind-restraint straps, welded strap connectors, and base flange attachment to roof structure at perimeter of curb, of size and spacing required to comply with wind-uplift requirements.
6. Platform Cap: Where portion of roof curb is not covered by equipment, provide weathertight platform cap formed from 3/4-inch- thick plywood covered with metal sheet of same type, thickness, and finish as required for curb.
7. Metal Counterflashing: Manufacturer's standard, removable, fabricated of same metal and finish as curb.

2.6 ROOF JACKS

A. Provide roof jacks acceptable to the roofing system manufacturer’s requirements and 20-year warranty.

2.7 MISCELLANEOUS MATERIALS

A. Provide materials and types of fasteners, protective coatings, sealants, and other miscellaneous items required by manufacturer for a complete installation.

B. Fasteners: Roof specialty manufacturer's recommended fasteners, designed to meet performance requirements, suitable for application and metals being fastened. Match finish of exposed fasteners with finish of material being fastened. Provide nonremovable fastener heads to exterior exposed fasteners. Furnish the following unless otherwise indicated:

1. Fasteners for Metallic-Coated Steel Sheet: Series 300 stainless steel or hot-dip zinc-coated steel in accordance with ASTM A153/A153M or ASTM F2329/F2329M.
2. Fasteners for Aluminum Sheet: Aluminum or Series 300 stainless steel.
3. Exposed Penetrating Fasteners: Gasketed screws with hex washer heads matching color of sheet metal.

C. Gaskets: Manufacturer's standard tubular or fingered design of neoprene, EPDM, PVC, or silicone or a flat design of foam rubber, sponge neoprene, or cork.

D. Elastomeric Sealant: ASTM C920, elastomeric polyurethane or silicone polymer sealant of type, grade, class, and use classifications required by roofing-specialty manufacturer for each application.

E. Butyl Sealant: ASTM C1311, single-component, solvent-release butyl rubber sealant; polyisobutylene plasticized; heavy bodied for hooked-type joints with limited movement.

F. Bituminous Coating: Cold-applied asphalt emulsion complying with ASTM D1187/D1187M.

G. Asphalt Roofing Cement: ASTM D4586, asbestos free, of consistency required for application.

2.8 GENERAL FINISH REQUIREMENTS

A. Comply with NAAMM/NOMMA AMP 500, "Metal Finishes Manual for Architectural and Metal Products," for recommendations for applying and designating finishes.

B. Protect mechanical and painted finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

C. Appearance of Finished Work: Noticeable variations in same piece are unacceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Install roof specialties in accordance with manufacturer's written instructions. Anchor roof specialties securely in place, with provisions for thermal and structural movement. Use fasteners, protective coatings, separators, underlayments, sealants, and other miscellaneous items as required to complete roof-specialty systems.

1. Install roof specialties level, plumb, true to line and elevation; with limited oil-canning and without warping, jogs in alignment, buckling, or tool marks.
2. Provide uniform, neat seams with minimum exposure of solder and sealant.
3. Install roof specialties to fit substrates and to result in weathertight performance. Verify shapes and dimensions of surfaces to be covered before manufacture.
4. Torch cutting of roof specialties is not permitted.

B. Metal Protection: Protect metals against galvanic action by separating dissimilar metals from contact with each other or with corrosive substrates by painting contact surfaces with coating or by other permanent separation as recommended by manufacturer's written installation instructions.
1. Only utilize coatings compatible with all materials they will be in contact with.

C. Expansion Provisions: Allow for thermal expansion of exposed roof specialties.

1. Space movement joints at a maximum of 12 ft. with no joints within 18 inches corners or intersections unless otherwise indicated on Drawings.
2. When ambient temperature at time of installation is between 40 and 70 deg F, set joint members for 50 percent movement each way. Adjust setting proportionately for installation at higher ambient temperatures.

D. Fastener Sizes: Use fasteners of sizes that penetrate wood blocking or sheathing not less than 1-1/4 inches for nails and not less than 3/4 inch for wood screws.

E. Seal concealed joints with butyl sealant as required by roof specialty manufacturer.

F. Seal joints as required for weathertight construction. Place sealant to be completely concealed in joint. Do not install sealants at temperatures below 40 deg F unless manufacturer approved for low temperature installation.

3.2 INSTALLATION OF ROOF CURBS:

A. Install each roof curb so top surface is level.

B. Seal and mechanically fasten extensions (where utilized) to existing curbs.

C. Adhere to the roofing system manufacturer's installation instructions for flashing.

D. Coordinate roof curb replacement with related electric, duct, fan re-installation.

3.3 INSTALLATION OF ROOF-EDGE DRAINAGE SYSTEMS

A. Install components to produce a complete roof-edge drainage system in accordance with manufacturer's written instructions. Coordinate installation of roof perimeter flashing with installation of roof-edge drainage system.

B. Gutters: Join and seal gutter lengths. Allow for thermal expansion. Attach gutters to firmly anchored gutter supports spaced not more than 24 inches apart. Attach ends with rivets and seal with sealant to make watertight. Slope to downspouts.

1. Install gutter with expansion joints at spacing not exceeding 50 ft. apart. Install expansion-joint caps.

C. Downspouts: Join sections with manufacturer's standard telescoping joints. Provide hangers with fasteners designed to hold downspouts securely to walls and 1 inch away from walls; locate fasteners at top and bottom and at approximately 60 inches o.c.

1. Provide elbows at base of downspouts at grade to direct water away from building when subdrains are not available.
2. Connect downspouts to underground drainage system indicated.

D. Conductor Heads: Anchor securely to wall with elevation of conductor top edge 1 inch below gutter discharge.

3.4 CLEANING AND PROTECTION

A. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing in accordance with ASTM A780/A780M.

B. Touch up factory-primed surfaces with compatible primer ready for field painting.

C. Clean and neutralize flux materials. Clean off excess solder and sealants.

D. Remove temporary protective coverings and strippable films as roof specialties are installed. On completion of installation, clean finished surfaces, including removing unused fasteners, metal

filings, pop rivet stems, and pieces of flashing. Maintain roof specialties in a clean condition during construction.

E. Replace roof specialties that have been damaged or that cannot be successfully repaired by finish touchup or similar minor repair procedures, as determined by COR.

END OF SECTION 077100